

Iterative design of a NAND hybrid riboswitch by deep batch Bayesian optimization

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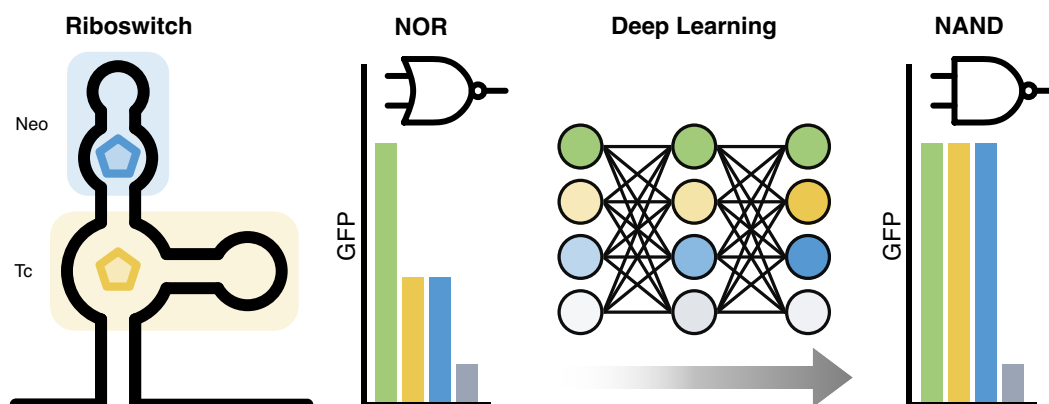
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Abstract

The design of large genetic circuits requires genetic regulatory devices capable of performing complex logic operations that place no excessive metabolic burden on the host cell. Hybrid riboswitches, synthetically enhanced compact RNA elements (<100 nucleotides) that form a tertiary structure with the ability to specifically bind two different target molecules, can be used to design genetic regulators that emulate Boolean logic. When inserted into the 5' UTR of a messenger RNA, these devices can regulate translation initiation upon specific binding of one or both ligands without the need for additional auxiliary factors. The goal of this study is to design hybrid riboswitches that emulate Boolean NAND logic in yeast. We propose a novel machine learning-based design framework combining high-throughput *in vivo* screening and deep Bayesian optimization. Through an initial screening, we discovered a hybrid riboswitch with NAND behavior. Using batch Bayesian optimization with an ensemble neural network as surrogate, we improved the NAND functionality of our hybrid riboswitch with respect to a performance score, thereby achieving near-digital NAND behavior. With its focus on model-based and score-driven design, our proposed method can complement experiment-driven approaches by allowing fine grained adaptation of functionality, including constructs sensitive to single nucleotide changes.

Graphical abstract



Introduction

Information processing plays a key role in cellular pathways as well as synthetically engineered devices. To engineer information processing synthetically, constructs imitating Boolean logic in cells are ideal [1–4]. RNA-based gene regulation,

as implemented by riboswitches, provides an appealing basis for the design of logic devices. Riboswitches are RNA sequences with the ability to influence gene expression through the formation of complex tertiary structures that specifically bind target molecules (ligands) [5, 6]. As they require no

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auxiliary factors to regulate gene expression, riboswitches impose only a low metabolic burden on the host organism [7]. While naturally occurring riboswitches have been discovered mostly in bacteria, synthetic riboswitches have been engineered to utilize novel regulatory mechanisms for implementation not only in bacteria, but also in eukaryotes [5, 8–11]. In *Saccharomyces cerevisiae*, synthetic riboswitches utilizing a roadblock mechanism prevent translation initiation by stabilizing their structure upon ligand binding [12, 13]. To enable the design of constructs implementing multi-input Boolean logic, riboswitch architectures capable of accepting two or more inputs are required. Hybrid riboswitches, which contain two different ligand binding pockets in one continuous structure, meet this requirement [14]. In addition to being excellent regulators, the ability to bind both ligands separately or simultaneously makes hybrid riboswitches suitable candidates for the implementation of Boolean logic. In particular, the hybrid riboswitch architecture has the potential to implement NAND behavior, a mechanistically challenging type of logic gate that has not been observed in any naturally occurring dual-input riboswitch [15]. Previous attempts at engineering riboswitches to implement NAND behavior have not taken advantage of direct interactions between different binding pockets. Instead, they have relied on a ribozyme to act as a shared expression platform that is controlled by two separately inserted binding pockets [16, 17].

The development of a synthetic riboswitch that utilizes the roadblock mechanism begins with an *in vitro* selection process known as SELEX (Systematic Evolution of Ligands by EXponential Enrichment) [18–22]. This method is used to discover sequences with high binding affinity for the chosen ligand, called aptamers. Subsequent *in vivo* screenings are required to identify candidates capable of undergoing conformational changes essential for mediating gene expression and to optimize their dynamic range. This approach focuses on dynamic range optimization based on a single input and is not suitable for the design of devices with multiple inputs (e.g. hybrid riboswitches). Previously reported hybrid riboswitches have been created using rational design [14], which poses a challenge when attempting to achieve specific switching behavior, due to the complex sequence-to-function relationship of the roadblock mechanism [23]. In addition, the sensitivity of (hybrid) riboswitches to single nucleotide changes makes the high-throughput generation of datasets containing regulatory active constructs for *in silico* approaches [24, 25] inefficient. This lack of robustness poses a challenge in the design of multi-input riboswitches that require structural, sequence-dependent interactions between the different ligand binding pockets to emulate a Boolean switching pattern. Additionally, the method has to cope with the exponential complexity of the sequence space.

In modern machine learning, Bayesian optimization [26–29], a sequential optimization method suitable for costly to evaluate black box functions, is often used in the context of hyper parameter optimization for deep learning [30–32], but has also been applied to biological engineering for the design of proteins [33–35], DNA [36], small molecules [37, 38], and experimental conditions [39–41]. In a round-based manner, new candidates for experimental evaluation are proposed and the gained information is taken into account in the next rounds. This compares to active learning [42–44], with the difference in the criterion for candidate selection. Us-

ing a probabilistic surrogate model that accounts for uncertainty, classically a Gaussian process [27, 45], Bayesian optimization can efficiently trade off exploration and exploitation, also in combinatorial design spaces such as RNA sequences. A promising trend—especially in life sciences—is the use of deep learning-based surrogate models [34–38], which are successfully applied as sequence-to-function models but require careful design for well-calibrated uncertainty quantification. For the candidate selection, different acquisition functions exist [26–29] and are often applied in a batch setting [27, 34, 37, 46], where multiple sequences are proposed at once. This increases experimental efficiency through parallel experimental characterization, but requires batch diversity to maximize the knowledge gained. Current methods maximize the joint uncertainty of the batch by assuming normally distributed predictions [37] or a heuristic selection of the top candidates according to a utility measure that does not account for interdependencies between candidates [34].

Following the appeal of riboswitches for information processing and the complexity of their design, the goal of this study is the engineering of digital-like NAND hybrid riboswitches through a novel machine-learning framework for sequence design. Precise emulation of NAND behavior by the hybrid riboswitch is required to ensure its viability in a wide range of applications. Essential to the design of this dual-input RNA device is a method actively coping with the sensitivity of hybrid riboswitches with respect to single nucleotide exchanges and explicitly considering the switching behavior to achieve. Specifically, the requirement for fine-tuning the four different states of the device to achieve near digital NAND behavior raises needs beyond those required for classical dynamic range maximization.

Therefore, we propose a multistep design approach incorporating *in vivo* screening and batch Bayesian optimization for score-driven sequence design. Our sequence design framework implements the batch acquisition function by using an ensemble neural network [47–50], which extends a custom pretrained transformer encoder [51, 52], as a surrogate and Kriging Believer (KB) [27, 46, 53, 54] as batching strategy. By considering batch diversity and allowing arbitrary distributions, our method goes beyond recent work. In addition, the network quantifies the certainty associated with its predictions and is thus applicable to low-throughput regimes such as those used to characterize hybrid riboswitches with high precision. Inspired by the design of hybrid riboswitches engineered for maximum dynamic range [14], we combined a tetracycline with a neomycin riboswitch and performed an *in vivo* screening to identify a variant showing NAND-like behavior, i.e. significant regulatory activity only in the presence of both ligands. This construct serves as input to our sequence design framework. The riboswitch design itself implements a design–build–test–learn (DBTL) cycle, in which the *in silico* candidate selection and the *in vivo* characterization alternate. The novel hybrid riboswitch we report here is capable of precisely emulating NAND logic through direct binding interactions, resulting in the stabilization of its compact structure only in the simultaneous presence of two different small molecule triggers. In addition, we present our versatile machine learning sequence design framework based on batch Bayesian optimization, which is adaptable to a wide range of design tasks, and provide insights into the method.

Methods

Experimental methods

Plasmid assembly

All constructs were cloned using the vector pCBB06, a variant of the plasmid pCBB05 containing no *gfp* start codon, as previously described [22]. Backbone linearization was performed immediately upstream of the *gfp* gene using the restriction sites for AgeI-HF and NheI-HF (NEB). Insert sequences containing a riboswitch sequence and start codon were generated through hybridization of oligonucleotides (Supplementary Table S1) with overhangs corresponding to the aforementioned restriction enzymes. Constructs were assembled through ligation (T4 DNA Ligase; NEB) and used for the transformation of *Escherichia coli* TOP10 (Invitrogen) following a standard 42°C heat shock protocol. The PureYield™ Plasmid Miniprep System (Promega) was used for plasmid extraction, with sequences being verified via Sanger Sequencing (Microsynth Seqlab).

Yeast cultivation and GFP measurements

Saccharomyces cerevisiae RS453α cells (MATα *ade2-1 trp1-1 can1-100 leu2-3 his3-1 ura3-52*) were transformed using the Frozen-EZ Yeast Transformation II Kit (Zymo Research). The cells were used for the inoculation of a 24-well plate, with each well (2× for each construct) containing 1.5 ml SCD-Ura [0.2% YNB w/o AA (Difco), 0.55% ammonium sulphate (Roth), 2% glucose (Roth), 12 µg/ml adenine (SIGMA), 1× MEM amino acids (SIGMA), 2% agar (Oxoid)]. After a 24-h incubation period at 30°C and 450 rpm, a 1:1000 dilution in fresh media containing either no ligand, 250 µM tetracycline or neomycin, or 250 µM of both ligands was performed in a new 24-well plate. After another 24 h of incubation the fluorescence of GFP and mCherry was measured for 25 000 cells per construct (20 µl culture + 180 µl SCD-Ura in a 96-well plate) using the CytoFlex S cytometer (Beckman Coulter) [488 nm laser (GFP); 561 nm laser (mCherry)]. Emission light was bandpass filtered at 510/520 and 610/620 nm. Cells containing the control plasmids (positive: pCBB05; negative: pCBB06) were treated equally to the constructs and measured in parallel. The GFP fluorescence of each candidate and control was normalized to its respective mCherry fluorescence and the values of the negative control were subtracted from all candidates and the positive control. The adjusted value of each sample was normalized to the positive control incubated with the equivalent ligand condition. A biological duplicate was measured for every candidate and every experiment was performed in technical triplicates at independent days. All bar plots present the mean of the technical triplicates. Statistical significance is assessed between lowest ON state and highest OFF state (score defining) via *T*-test assuming identical variance. *P*-values are reported in respective figures (n.s. = nonsignificant = $p > .05$, * = $.01 < p \leq .05$, ** = $.001 < p \leq .01$, *** = $p \leq .001$).

Yeast library pool generation

Hybrid riboswitch DNA pools containing randomized regions (Supplementary Table S2) were amplified (Q5 High-Fidelity DNA polymerase; NEB) to add regions homologous to the insertion site on the plasmid pCBB06. *Saccharomyces cerevisiae* RS453α cells were transformed according to a high-efficiency electroporation protocol [55], using 1.5 µg linearized plasmid pCBB06 (AgeI-HF and NheI-HF) and 1.5 µg amplified pool

DNA, as previously described [14]. Prior to transformation, pool and plasmid DNA were purified using the Wizard SV Gel and PCR Clean-Up System (Promega). Two transformations were performed for each pool to increase transformation efficiency, and the transformed cells were combined in 250 ml SCD-Ura in a 2 l culture flask and incubated for 48 h at 30°C and 110 rpm. After the initial 48-h incubation period, serial dilution steps of the liquid culture in 250 ml SCD-Ura were performed according to the electroporation protocol to avoid the occurrence of double-transformed cells (1/10, 1/30, 1/100; 24-h incubation after each dilution).

NAND communication module screening

A yeast library culture (see above; sequence pool TNNAND1_N6) was used to inoculate 4 × 250 ml SCD-Ura cultures (1:100; 2 l flasks). Each inoculated culture contained a different ligand condition (w/o; 250 µM tetracycline; 250 µM neomycin; 250 µM both). Positive (pCBB05) and negative (pCBB05) controls were generated by transforming *S. cerevisiae* RS453 cells with the respective plasmids using the Frozen-EZ Yeast Transformation II Kit (Zymo Research). Cultures were incubated at 30°C for 24 h in a humidified incubator (shaking). Prior to sorting, cells were diluted (1:10) in 1× phosphate-buffered saline. For fluorescence-activated cell sorting the SONY SH800 cell sorter was used [488 nm laser (GFP), bandpass filter: 525/550 nm]. Doublets (connected cells) and cells of irregular shape or size were excluded. The GFP fluorescence range (0%–100%) for cell sorting was defined as the distance between the peaks of the controls. The fluorescence range was divided into eight equally sized gates on a linear scale, with two additional gates covering the fluorescence ranges occupied by the controls. For each ligand condition, 12 500 candidates per gate were sorted into a separate well (500 µl SCD-Ura; 1 ml SCD-Ura added after sorting) on a 24-well plate. The plates were sealed (Greiner Bio-One Breathseal membrane) and incubated for 48 h at 30°C using a plate incubator. Afterward, plasmid DNA was extracted from all wells using the Zymoprep™ Yeast Plasmid Miniprep II kit (Zymo Research) according to the manufacturer's instructions. The sequence region containing the partially randomized hybrid riboswitch was amplified and the PCR product was purified using the Wizard® SV Gel and PCR Clean-Up System (Promega) according to the manufacturer's instructions. The sequence content of all samples was analyzed using NGS (AMPLICON-EZ, GENEWIZ). As the gates span the whole fluorescence range and do not overlap, we approximate each sequence's fluorescence by the frequency-weighted average of the fluorescence intensities assigned to each gate. This is done for all four ligand conditions, whereby the gate's fluorescence intensity is defined as the average fluorescence of events in a gate.

NAND hybrid riboswitch dynamic range screening

Hybrid riboswitch yeast libraries for the sequence pools TNNAND8_T1(N10)/T2(N10)/T3(N10)/T4(N10) were created (see Yeast Library Pool Generation) and combined into one culture (1 ml of each library added to one 100 ml SCD-Ura culture). Positive (pCBB05) and negative (pCBB05) controls were generated as previously described (see the “NAND communication module screening” section). All cultures were incubated for 24 h at 30°C in a humidified incubator (shaking). The combined library and the controls were each used to inoculate (1:100) two 10 ml SCD-Ura cultures,

containing either no ligand or 250 μM tetracycline and neomycin, and again incubated for 24 h at 30°C in a humidified incubator (shaking). Prior to sorting, cells were diluted (1:10) in 10 ml fresh SCD-Ura. For fluorescence-activated cell sorting, the CytoFlex SRT cell sorter (Beckman Coulter) was used [488 nm laser (GFP); 561 nm laser (mCherry); band-pass filter: 525/540 and 610/620 nm]. Doublets (connected cells), cells of irregular shape or size and cells lacking constitutive mCherry expression were excluded. GFP fluorescence was normalized to mCherry fluorescence and the fluorescence range (0%–100%) for cell sorting was defined as the distance between the peaks of the controls. In the first sorting step, 10% of all cells exhibited low fluorescence (0%–10%) in the presence of both ligands. 1.2 million cells were sorted into 5 ml fresh SCD-Ura medium. The sorted cells were incubated for 48 h at 30°C in a humidified incubator (shaking). Afterward, two new 10 ml cultures, one for each ligand condition (w/o; 250 μM tetracycline and neomycin), were inoculated (1:100) and incubated for 24 h as previously described. The high-fluorescence area was divided into four gates (20%–40%, 40%–60%, 60%–80%, 80%–100%) for the second sorting step, in case NAND behavior would begin to be compromised at higher expression levels. For each gate the amount of sorted cells was chosen based on the percentage of the population it contained to guarantee complete coverage (20%–40% GFP: 360k cells; 40%–60% GFP: 36k cells; 60%–80% GFP: 10k cells; 80%–100% GFP: 10k cells). Cells were sorted into four separate 5 ml SCD-Ura cultures and incubated for 24 h as previously described. A final sorting step was performed separately for each subpopulation in the presence of both ligands as previously described (0%–10% fluorescence; 10k cells sorted each) to further enrich candidates with low OFF state fluorescence. Plasmid DNA was extracted from all four cultures as previously described (see the “NAND communication module screening” section). The sequence content of all samples was analyzed using NGS (AMPLICON-EZ, GENEWIZ).

In vitro transcription

RNA used for *in vitro* analysis [isothermal titration calorimetry (ITC), in-line probing] was *in vitro* transcribed from linear double-stranded DNA (dsDNA) templates. Each template contained a T7 promoter, followed by 3× guanosyl residues for transcription initiation upstream of the RNA construct sequence. Templates were ordered as single-stranded DNA (ssDNA) oligonucleotides with a 20 nucleotide overlap and amplified using a Q5[®] High-Fidelity DNA polymerase (NEB) according to the supplier’s instructions. The resulting dsDNA was purified through a phenol/chloroform extraction, followed by ethanol precipitation, as previously described [56]. The *in vitro* transcription reaction containing the purified dsDNA template (50 $\mu\text{g}/\text{ml}$), 20 mM Mg(Ac)₂, 200 mM Tris-HCl (pH 8.0), 20 mM DTT, 2 mM spermidine, 4 mM of each NTP, 15% DMSO, and 1000 U/ml T7 polymerase (self-made) was incubated for 6 h at 37°C. The resulting RNA of correct size was purified via excision from a polyacrylamide gel. RNA concentration was determined using a NanoPhotometer[®] (IMPLEN).

Isothermal titration calorimetry

ITC measurements of RNA constructs were carried out in binding buffer I (20 mM Tris-HCl, pH 7.5, 10 mM MgCl₂, 100 mM NaCl) [57]. For each measurement, a 10 μM RNA solution and a separate 100 μM ligand solution (tetracycline

or neomycin) were prepared in the same buffer. Experiments were performed using a MicroCal PEAQ-ITC (Malvern Panalytical, UK) containing buffered RNA in the sample cell (200 μl) and ligand solution in the syringe (40 μl). After equilibration at 25°C and a 60 s delay, a test injection of 0.2 μl in 0.4 s was performed. For the measurement, 13 injections of 2 μl in 4 s were carried out in 150-s intervals at a stirring speed of 750 rpm. Measurement results were recorded as differential power ($\mu\text{cal}/\text{s}$) over time (min). The heat associated with each titration peak was integrated and plotted against the molar ratio of ligand and RNA. The dissociation constant (K_d) was determined from the fitted binding curve using the MicroCal PEAQ-ITC Analysis Software (version 1.41). After titration of the first ligand (tetracycline or neomycin), the same RNA sample was titrated with the second ligand to measure binding affinity for the second ligand in the presence of the first. A replicate was performed for each ITC measurement.

In-line probing

RNA dephosphorylation was carried out using an alkaline phosphatase (Roche) (15 min at 50°C), followed by a heat inactivation step (3 min at 95°C). Afterward, radioactive 5′ labeling was performed with [γ -³²P]-ATP (6000 Ci/mmol, 10 mCi/ml; HARTMANN ANALYTIC) using a T4 polynucleotide kinase (Roche) (40 min at 37°C). The ³²P-labeled RNA was purified via gel excision from a polyacrylamide gel. Separate in-line reactions were prepared for all ligand conditions (w/o, neomycin, tetracycline, both), each containing 25 kCPM of ³²P-labeled RNA, In-Line reaction buffer (10 mM Tris-HCl, pH 8.3, 10 mM MgCl₂, and 100 mM KCl) and 1 mM of (each) ligand. After incubation at room temperature for 72 h, reactions were stopped by adding an equal amount of 2× denaturing RNA loading buffer [deionized formamide, 25 mM ethylenediaminetetraacetic acid (EDTA), bromophenol blue). ³²P-labeled RNA was prepared as a size marker through partial alkaline digestion in 10× Na₂CO₃ buffer (0.5 M Na₂CO₃, 10 mM EDTA) for 7 min at 95°C. For guanine residue identification a separate reaction containing ³²P-labeled RNA and 1 U of T1 RNase was incubated for 5 min at 55°C. Untreated ³²P-labeled RNA was prepared as a control. All samples were mixed with 2× denaturing RNA loading buffer and loaded on a denaturing polyacrylamide gel. After electrophoresis, the gel was dried on a heated vacuum bed overnight and analyzed via phosphorimaging (Amersham Typhoon, GE Healthcare).

Machine learning methods

Bayesian optimization

Bayesian optimization is a global optimization method using a surrogate model, in our case a deep ensemble neural network, to guide optimization. The surrogate model describes the belief $p(S(x) | D_j)$, the distribution over $S(x)$ conditioned on the knowledge D_j in round j . This knowledge is

$$D_j = \{(x, s) : s = S(x), x \in \bar{\mathcal{X}}_{j-1}\}$$

where $\bar{\mathcal{X}}_{j-1}$ is the set of sequences experimentally characterized in the previous rounds.

Derived from the model’s belief, the acquisition function assigns a utility value $u(x)$ to each sequence x and selects the candidate by solving $x' = \arg \max_{x \in \mathcal{X}} u(x)$.

This optimization problem can be solved by exhaustive enumeration, stochastic optimization or gradient based approaches exploiting differentiability of $u(\cdot)$. Enabled by the compact size of the final design space considered here, we solve this by exhaustive enumeration of the candidate sequence pool.

Kriging Believer

Kriging Believer (KB) is a natural extension of Bayesian optimization to the batch setting, as it replaces costly function evaluations by the model's predictions and otherwise sticks to the sequential decision process. For iteration i of batch Bayesian optimization round j , the knowledge is

$$\overline{D}_j^{(i)} = D_j \cup \{(x, \mu(x)) : x \in \{x'_1, \dots, x'_{(i-1)}\}\},$$

which is the union of D_j , the actual measurements at the start of round j , and the augmented dataset given by the predictions $\mu(x'_k) = E[S(x'_k) | \overline{D}_j^{(k)}]$ for the sequences $\{x'_1, \dots, x'_{(i-1)}\}$ already part of the batch. By training the ensemble model from scratch (except the pretrained encoder), the model's belief is updated and explicitly conditions the predictions on the augmented values (i.e. $p[S(x) | \overline{D}_j^{(i)}]$). The batch of sequences $x'_1, \dots, x'_m$ is then experimentally characterized and the knowledge for the next round of batch Bayesian optimization is

$$D_{j+1} = D_j \cup \{(x, s) : s = S(x), x \in \{x'_1, \dots, x'_m\}\}.$$

Encoder model—architecture and training

The encoder model follows a BERT-like transformer encoder architecture [51, 52] including multihead attention [51]. For tokenization, we build upon the 3-mer tokenizer of DNABERT [58], representing the construct sequence by 3-mer tokens (e.g. [AGC]). Besides, we use [CLS] as start token, [SEP] as end token, and [PAD] as padding token. The combination of [CLS], [SEP], and [PAD] tokens allows to have inputs of varying length.

The encoder implementation consists of an embedding layer, sinusoidal positional encoding [51], and six transformer encoder blocks. The embedding layer turns tokens into vectors of dimension $d_{\text{model}} = 128$. Each transformer encoder includes a multihead self-attention mechanism consisting of four heads ($b = 4$) with dimension $d_{\text{attention}} = 64$. In comparison to the original work of Vaswani *et al.* [51], where $d_{\text{attention}} = d_{\text{model}}/b$ for reasons of computational cost, we increase the attention dimension by a factor of two due to the smaller embedding dimension d_{model} (i.e. in our case $d_{\text{attention}} = 2 d_{\text{model}}/b$). The two layer position-wise feed-forward neural network is applied to each token position and has input and output dimension $d_{\text{model}} = 128$ and hidden dimension 384. Following Devlin *et al.* [52], the first layer employs the Gaussian error linear units (GELU) and the second a linear activation function. Both layers feature a dropout probability of 10%. The remainder of the transformer encoder follows the design of Vaswani *et al.* [51].

Our encoder model is trained on 36 925 unique riboswitch sequences obtained from sequencing. We apply train (70%), validation (15%), and test set (15%) splits. The considered riboswitch constructs have sequence lengths of up to 70 bases. After 3-mer tokenization, the maximum length is 68 tokens and together with mandatory start and end tokens the maximum token sequence length is 70. Token sequences shorter

than that are extended with the [PAD] token. The training was carried out on a consumer grade NVIDIA GPU with a batch size of 64. Following four warmup epochs, the learning rate was determined by an exponential learning rate schedule with an initial learning rate of $8 \cdot 10^{-4}$ and $\gamma = 0.98$. The learning rate is updated after each epoch and early stopping stops training after multiple iterations without improvements. This pretraining uses the loss defined by Equation (2) and includes the masked language model (MLM) approach [52] and the triplet margin loss [59].

For the MLM objective, we employ online masking [60] and otherwise follow the prescriptions of Devlin *et al.* [52]. In particular, 15% of the tokens in the token sequence (the special tokens [CLS], [SEP], and [PAD] are not considered) are selected randomly to be masked. From the tokens to mask, 80% will be replaced by the [MASK] token, 10% by a random 3-mer token, and 10% are not altered. For the masked token prediction, the encoder is extended by a position-wise single-layer feed-forward network predicting the tokens at the respective positions. The MLM loss is then the cross-entropy loss of the predictions and the actual tokens, whereby only the masked positions are taken into account.

The triplet margin loss [59] focuses on the quality of the embedding and requires an anchor, a positive and a negative. We here use the output of the [CLS] token as sequence embedding (yields embedding vectors of size $d_{\text{model}} = 128$). As anchors, we use the unmasked token sequences in the batch. The positive is obtained by applying masking to the anchor and the negative is the masked version of a token sequence identified via hard negative selection [59]. This selection criterion computes the embedding distance between each anchor in the batch and all other sequences and selects the one with the smallest nonzero distance. We then apply the triplet margin loss with a margin of $\gamma = 1.0$ to the embeddings derived by the sequence encoder.

The training loss is calculated as the superposition of the MLM loss and the triplet margin loss. The best parameter configuration is selected based on the evaluation on the validation dataset and the final model achieved an accuracy of 96.9% on the test dataset.

Ensemble and regression model—architecture and training

The regression model is a fully connected multilayer perceptron using the sequence embeddings as input and outputting the expression levels corresponding to the four riboswitch conformations. It consists of nine intermediate layers, each of dimension 256, and a linear output layer. GELU is used as the activation function and no dropout is used. To improve training performance, we use layer normalization after every three hidden layers.

Joining the pretrained sequence encoder and the regression model yields the sequence-to-function neural network. $n = 100$ instantiations of this network constitute the ensemble model. To achieve diversity among models in the ensemble, the multilayer perceptrons are initialized randomly and the training procedure is designed to prevent ensemble collapse. In particular, dropout is only used within the encoder, as initial evaluations showed that further use of dropout negatively affects the network's ability to differentiate known and unknown sequences. As hyperparameters we used a batch size of 4, 120 epochs, a learning rate of 10^{-4} , a learning rate decay of 0.99, 4 initial warmup epochs, and a train validation split of 75% and 25%. The training includes the training of

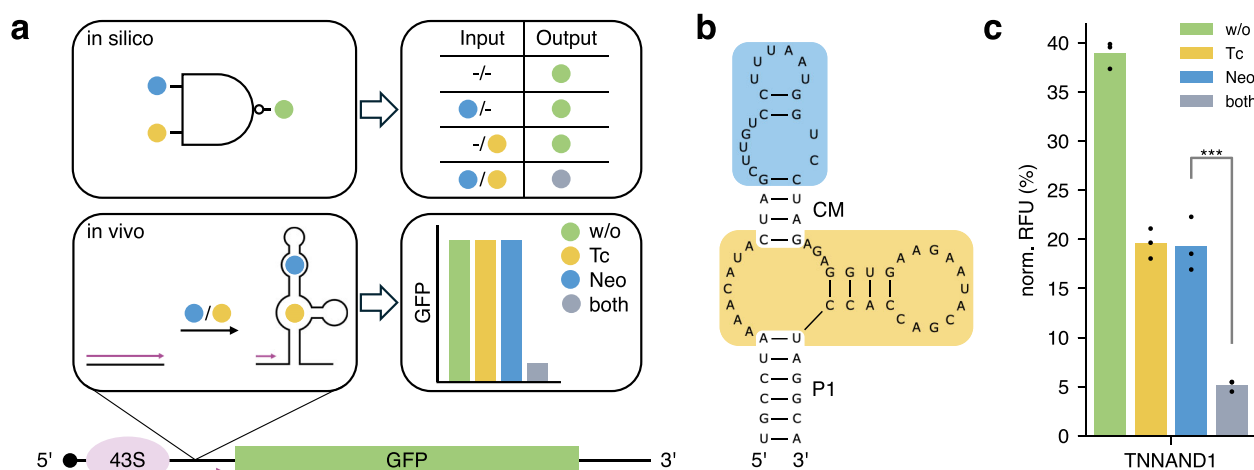


Figure 1. Translation regulation in yeast by a tetracycline–neomycin hybrid riboswitch following Boolean logic. **(a)** Schematic representation of the roadblock mechanism of translation regulation in yeast being performed by a tetracycline–neomycin hybrid riboswitch following NAND logic (blue: neomycin, yellow: tetracycline). **(b)** 2D structure prediction of the TNNAND1 hybrid riboswitch. The binding pockets for neomycin and tetracycline have been highlighted in blue and yellow, respectively. **(c)** Effect of TNNAND1 on GFP expression in the absence (w/o) and the presence of 250 μM tetracycline (Tc), neomycin (Neo), or both. *Saccharomyces cerevisiae* RS453 were transformed with the plasmid pcBB06 containing TNNAND1 in the 5' UTR of a constitutively expressed *gfp* gene. Cells were incubated under the respective ligand conditions for 24 h. GFP values were normalized to constitutively expressed mCherry to eliminate cell-to-cell expression variability. pcBB06 without an insert was used as the blank, and its values were subtracted from all other measurements. The fluorescence values of each measurement were normalized to their respective ligand-specific positive control (pcBB05). All measurements were performed in duplicates and repeated twice on separate days. *P*-Values corresponding to worst-case dynamic range are indicated (see the “Methods” section).

the sequence encoder. Each model in the ensemble is trained individually on the same dataset.

As loss function for the regression task we use the Kullback–Leibler divergence (KL divergence) [61]. In particular, we consider the individual expression levels predicted—which are in the range between 0 and 1—as Bernoulli probabilities, characterizing the distribution of riboswitches expressing proteins or not. This Bernoulli KL divergence for two Bernoulli distributions characterized by the probabilities p (the predicted value) and q (the target value) is given by

$$D_{\text{KL}}(p||q) = \log\left(\frac{1-p}{1-q}\right) + p \log\left(\frac{p}{q}\right).$$

The KL divergence measures the nonmetric distance between two distributions and as such is minimal in case of identical distributions.

Results

Hybrid riboswitch design

In a first step, we screened for a hybrid riboswitch that implements Boolean NAND-like behavior, exhibiting significant inhibition of gene expression only in the simultaneous presence of both ligands, while maintaining equally high expression levels in the absence of either or both ligands (Fig. 1a). We combined two well-established riboswitches that recognize tetracycline [13] and neomycin [62], respectively, by replacing the P2 stem of the tetracycline riboswitch with a truncated version of the neomycin riboswitch (Fig. 1b). We refer to the stem sequence connecting the two binding pockets as a communication module. We designed the hybrid riboswitch to contain a short communication module that enforces close spatial proximity between the two binding pockets, thus allowing direct interactions. In addition, we modified the closing P1 stem, as it was shown that expression levels regulated by tetracycline-responsive hybrid riboswitches can be modulated

when using different versions of P1 [14]. This effect was previously observed for the original tetracycline riboswitch [13, 63] (Supplementary Fig. S1). We used the tetracycline–neomycin variant TN6 [14] as the starting point, as it exhibits only moderate dynamic ranges in the individual presence of either ligand. We considered this behavior to be a favorable trait for an initial NAND screening template. An additional UA base pair was added to the base of the P1 stem to increase the overall stability of the construct. The sequence of the resulting construct TNNAND1 is depicted in Fig. 1b. The effect of TNNAND1 on gene expression was analyzed by inserting the construct into the 5' UTR of a *gfp* reporter gene directly upstream of the start codon. GFP expression was measured after a 24-h incubation period in the absence and the presence of either or both ligands [w/o; 250 μM tetracycline (Tc); 250 μM neomycin (Neo); 250 μM both]. The dynamic range achieved by TNNAND1 in the dual-input state is 7.6-fold, while remaining 2-fold in the presence of only one ligand (Fig. 1c). While the deviation between the state without ligand and the individual ligand states means that TNNAND1 does not achieve any defined logic behavior, we found the similarity of the two individual ligand states to be a good starting point for further engineering.

Screening for a NAND communication module

We designed a library of different riboswitch sequences through randomization of the TNNAND1 communication module between the binding pockets for tetracycline and neomycin, except for the conserved closing GC base pair of the internal bulge of the neomycin binding pocket, resulting in an N6 pool. For the initial *in vivo* screening, we sorted yeast cells containing the different N6 variants based on their GFP fluorescence. However, sorting specifically for high fluorescence in the presence of one or no ligand and low

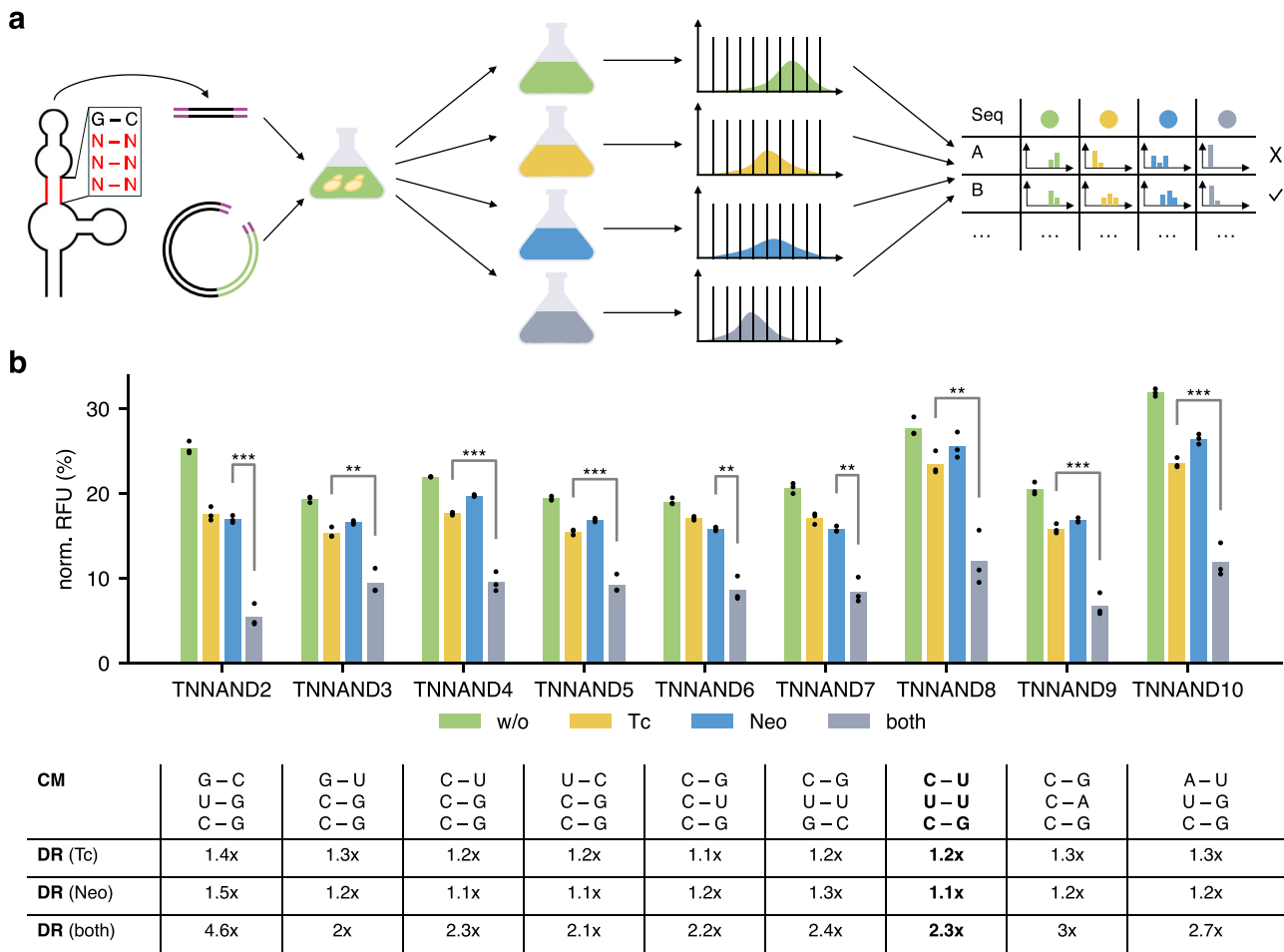


Figure 2. NAND communication module screening top candidates. **(a)** Screening workflow. A yeast library containing the TNNAND1 sequence with a randomized communication module sequence was created via homologous recombination. Cultures for each ligand condition were inoculated and sorted through the same grid of equally sized gates. The plasmid DNA from each subpopulation was extracted and analyzed via NGS. The resulting datasets were used to approximate the switching pattern of each sequence discovered under all four ligand conditions. **(b)** The influence on GFP expression of candidates that emulate NAND behavior was measured in the absence (w/o) and the presence of 250 μ M tetracycline (Tc), neomycin (Neo), or both. *Saccharomyces cerevisiae* RS453 were transformed with the plasmid pcBB06 containing the respective candidate in the 5' UTR of a constitutively expressed gfp gene. Cells were incubated under the respective ligand conditions for 24 h. GFP values were normalized to constitutively expressed mCherry to eliminate cell-to-cell expression variability. pcBB06 without an insert was used as the blank and its values were subtracted from all other measurements. The fluorescence values of each measurement were normalized to its respective ligand-specific positive control (pcBB05). All measurements were performed in duplicates and repeated twice on separate days. The communication module sequence of each candidate is depicted below its respective measurement results. *P*-Values corresponding to worst-case dynamic range are indicated (see the "Methods" section).

fluorescence in the presence of both ligands was not sufficient to guarantee the discovery of candidates with NAND behavior. This is because all variants contained in the N6 pool have their own sequence-specific expression levels, and candidates with the most NAND-like behavior are not guaranteed to also achieve high expression levels in the absence of both ligands or a high dynamic range in their presence, making a standard sorting approach difficult. To circumvent this issue, we divided the maximum GFP fluorescence range (0%–100%) into subpopulations using equally sized gates, thereby creating a sorting grid that we could use for all ligand conditions. We were able to determine the exact sequences contained in each subpopulation using next-generation sequencing (NGS). Because we used the same grid for all conditions, subpopulations originating from a specific gate always contained candidates corresponding to its fluorescence range. This strategy allowed us to approximate the switching pattern of each sequence contained in the N6 pool by compar-

ing its occurrence in different subpopulations for all ligand conditions.

We determined the maximum GFP fluorescence range using control plasmids and divided it into eight equally sized gates on a linear scale ranging from low to high fluorescence. Two additional outer gates cover the fluorescence ranges occupied by the controls. We then used a yeast library containing the N6 pool to inoculate four different cultures, one for each ligand condition (w/o; 250 μ M Tc; 250 μ M Neo; 250 μ M both). Using the predetermined grid of gates, we divided each culture into subpopulations. We sorted 12 500 candidates per gate for each ligand condition. Plasmid DNA was extracted from each subpopulation and the contained sequences were verified using NGS. We then approximated the switching pattern of each sequence by comparing its occurrence in different subpopulations for every ligand condition, with subpopulations originating from gates in the lower fluorescence range corresponding to low expression levels and vice versa (Fig. 2a).

We identified sequences occurring in lower gates for the double ligand condition, while showing similar distributions across gates representing higher fluorescence for all other ligand conditions using the NGS datasets. We then made an expert selection of sequences that demonstrated the most promising NAND-like behavior based on their overall performance across all ligand conditions. We proceeded to clone the chosen candidates TNNAND2-10 and individually measure their effect on GFP expression. The communication module sequences of TNNAND2-10, as well as their dynamic ranges for each ligand condition are depicted in Fig. 2b. All candidates present NAND logic to some degree, with dynamic ranges for the individual ligand conditions remaining low in comparison to the double ligand condition (e.g. TNNAND2: Tc = 1.4×, Neo = 1.5×, both = 4.6×). However, constructs with a higher dynamic range in the presence of both ligands also present a stronger decrease in GFP values for the individual ligand conditions. Constructs with a low dynamic range in the presence of either ligand suffer from poor regulation, even in the presence of both ligands (e.g. TNNAND8: Tc = 1.2×, Neo = 1.1×, both = 2.3×). We identified TNNAND8 as the most interesting candidate for further optimization. Given that the sequence of the tetracycline riboswitch closing stem P1 strongly influences switching behavior [14, 63], our next step was to engineer a TNNAND8-specific P1 stem to improve dynamic range while maintaining the NAND behavior conferred by the new communication module.

Instead of six nucleotides (three base pair long communication module), which yield $4^6 = 4096$ combinations in total, the P1 stem region includes five or even more base pairs, increasing the combinatorial search space to over one million combinations. Due to this exponential increase in complexity, the characterization of expression levels by cloning and screening sequence pools of this size and variety becomes challenging. In particular, this requires a significant increase in the number of cells sorted per fluorescence gate to maximize the pool coverage and preserve the desired precision. However, the increase of time needed for sorting also poses challenges on the accuracy, as the measurements have to match the timings of the experimental protocols.

Bayesian optimization to navigate the sequence space

The combinatorial increase in complexity renders purely experimental-based sequence design inefficient and we thus next investigated the use of Bayesian optimization in combination with sequence-to-function surrogate models for efficient and functionality-driven riboswitch stem design. Incorporating individual construct measurements, the learned sequence-to-function model can assess the functionality of yet unexplored riboswitch sequences, while Bayesian optimization for sequence selection efficiently trades off exploration and exploitation to cope with the combinatorial complexity of the sequence space.

The P1 stem design is a particular instance of biological sequence design, i.e. the task of assigning nucleotides in an engineered construct to achieve a desired behavior. This behavior can be specific expression levels, dynamic range maximization, particular sequence features or—as in our case—the emulation of a particular Boolean function. We assess the degree of functionality via a scoring function $S(\cdot)$, which is based on experimental characterization. Consequently, we formalize

the design task in terms of an optimization problem with the ideal design given by

$$\hat{x} = \arg \max_{x \in \mathcal{X}} S(x),$$

where $\mathcal{X}$ is the sequence design space and $\hat{x}$ maximizes the score $S(\cdot)$. To solve this design task, we build a sequence design framework on the basis of Bayesian optimization.

Core to Bayesian optimization is the model's belief $p[S(x) | D_j]$, expressed as a probability distribution over scores conditioned on the knowledge or data D_j available at optimization round j . For a particular sequence $x \in \mathcal{X}$, the belief represents a probabilistic sequence-to-function mapping giving rise to not only the expected function value in terms of $E[S(x') | D_j]$ but also the associated uncertainty. This serves as a surrogate to guide Bayesian optimization also in regions not yet explored by experimental assessment. In particular, the acquisition function, which determines the candidates to evaluate next, selects the sequence x maximizing the utility function $u(x)$. This utility function is derived from the probabilistic predictions of the surrogate model.

We select the upper confidence bound (UCB) [27, 28, 29, 46, 53, 64, 65] as the acquisition function for the sequence design task because of interpretability, expressiveness, and simplicity of evaluation in combination with deep ensemble models. The UCB estimates the utility of each sequence by an upper bound on the achievable score. Formally, the utility function $u(x)$ is defined to satisfy

$$P[S(x) \leq u(x) | D_j] = 1 - \alpha, \quad (1)$$

where $P[S(x) \leq s | D_j]$ is the cumulative distribution function corresponding to $p[S(x) | D_j]$ and $\alpha \in [0, 1]$ defines the desired level of confidence and can be used to trade off exploitation ($\alpha \rightarrow 0.5$) and exploration ($\alpha \rightarrow 0$). For example, $\alpha = 0.05$ leads to a utility $u(x)$ such that, according to the model, the probability of $S(x)$ exceeding $u(x)$ is at most 5%. The ensemble model as surrogate allows us to define the UCB directly in terms of Equation (1) by using the quantiles of the empirical distribution, i.e.

$$u(x) = q_{1-\alpha}(S(x)),$$

where $q_{1-\alpha}(S(x))$ refers to the $1 - \alpha$ quantile function. The sequence x' maximizing $u(\cdot)$ is then chosen to be evaluated experimentally (see Fig. 3a).

Experimental efficiency through batch creation

We devise a novel Bayesian optimization method for the batch setting [27, 46, 53] to make use of the parallelism of the assay characterizing the riboswitches, where multiple candidate sequences $x'_1, \dots, x'_m$ need to be proposed at once. Repeatedly querying the acquisition function without updating the model's belief will either (1) select the same riboswitch or, by excluding case (1) explicitly, (2) select sequences with high proximity in terms of the score $S(\cdot)$. The consequence of both cases is the absence of diversity, again being detrimental to experimental efficiency.

We achieve batch diversity by extending our proposal mechanism by the KB technique [27, 46, 53, 54], explicitly excluding case (1). The idea of KB is to skip the experimental evaluation and instead update the model's belief with the predicted values. As such, the uncertainty vanishes for candidates in the current batch and decreases in their vicinity. Upon selecting the m candidates, the batch $x'_1, \dots, x'_m$ is

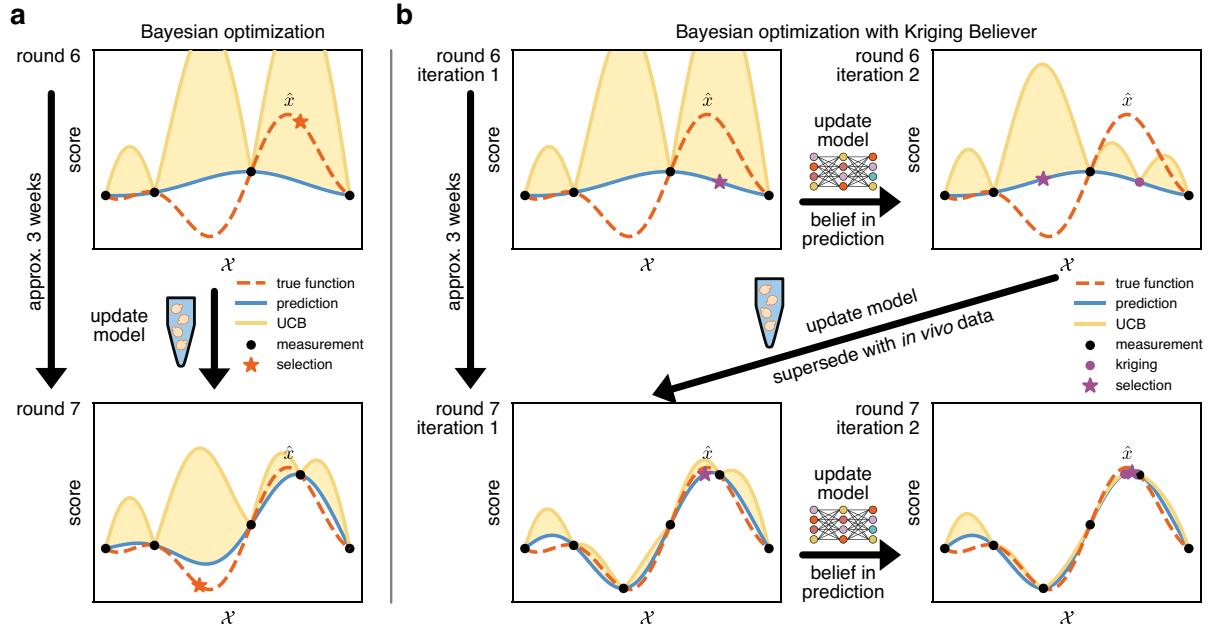


Figure 3. Batch Bayesian optimization. Side-by-side comparison of Bayesian optimization (a) and batch Bayesian optimization employing KB (b). (a) Bayesian optimization proposes a single sequence (orange star) maximizing the utility function (here UCB, yellow line) per round. The surrogate model’s belief (blue line and UCB) is updated with the *in vivo* sequence characterization. (b) KB introduces multiple iterations per batch Bayesian optimization round and also selects the sequence maximizing the utility function. Between iterations, the model’s belief is updated with the predicted value (purple star). Upon selection of all sequences for the current batch (in this example $m = 2$), the model’s belief is updated with the *in vivo* characterization of the batch. Errors introduced by kriging are superseded between rounds (compare round 6 iteration 2 and round 7 iteration 1).

experimentally characterized and the model is updated with the true measurements. Figure 3b visualizes this technique for $m = 2$, allowing a side-by-side comparison of Bayesian optimization and KB based batch Bayesian optimization. For clarity, we refer to a round as a single run of batch Bayesian optimization and to an iteration as a single step of Bayesian optimization as part of a batch Bayesian optimization round (compare Fig. 3a and b).

While KB blends nicely with Bayesian optimization and provides an appealing, intuitive, and straightforward approach for the extension to the batch setting, the approach may raise concerns regarding error propagation. Every evaluation of the utility function—with the exception of the first iteration—explicitly depends on the predictions made in the previous iterations (see Fig. 3b round 6, iteration 2). These predictions are usually not exact and prediction errors can negatively affect future predictions. However, these errors do not propagate into the next round of batch Bayesian optimization, as all predictions are superseded by the experimental characterization (compare Fig. 3b round 6, iteration 2 to round 7, iteration 1). Besides, future rounds of batch Bayesian optimization as well as the uncertainty reduction in the proximity of sequences in the batch are independent of the particular predictions (see Fig. 3b round 6, iteration 1). As such, beneficial candidate sequences previously discarded because of incorrect predictions can be reconsidered later.

Deep ensemble neural network for uncertainty quantification

We propose the use of an ensemble neural network [47–50] representing a probabilistic sequence-to-function mapping to guide the Bayesian optimization in the sequence space. By learning the sequence-to-function relations from experimen-

tal data, this model allows to assess yet uncharacterized sequences.

We create each of the n networks in the ensemble from a sequence encoder featuring a specifically designed BERT-like transformer encoder [51, 52] and a fully connected regression network (MLP) [66]. To train the ensemble, we employ a two stage training process. First, we pretrain the sequence encoder on a task specific sequence dataset to learn the structure inherent to the sequence space under consideration. The masked language model (MLM) technique [52], which is widely used in natural language processing, learns in-sequence relationships by randomly masking parts of a sequence and requiring the model to predict the masked parts. As previous studies have shown the minor quality of embeddings learned by relying on MLM alone [67], we extend our pretraining with a novel metric learning task that we derived from the triplet margin loss used in classification tasks [59, 68, 69].

As depicted in Fig. 4a, we combine both objectives by forming a triplet network of identical sequence encoders. During pretraining, we first select sequence x_1 and a corresponding hard negative x_2 (see the “Methods” section). From these, we derive the anchor $x_a = x_1$, the positive $x_p = \text{mask}(x_1)$, and the negative $x_n = \text{mask}(x_2)$ by masking, as described previously [52] and outlined in the “Methods” section. The triplet network computes the corresponding embeddings v_a , v_p , v_n and predicts the masked parts of x_p and x_n . Our overall training loss is then defined as

$$\mathcal{L} = \mathcal{L}_{\text{MLM}} + \mathcal{L}_E, \quad (2)$$

where $\mathcal{L}_{\text{MLM}}$ is the MLM loss [52] and $\mathcal{L}_E$ is the triplet margin loss [59, 68, 69] with margin parameter γ and euclidean norm $\|\cdot\|_2$ (see Fig. 4b).

$$\mathcal{L}_E = L(v_a, v_p, v_n) = \max[0, \|v_p - v_a\|_2 + \gamma - \|v_n - v_a\|_2]$$

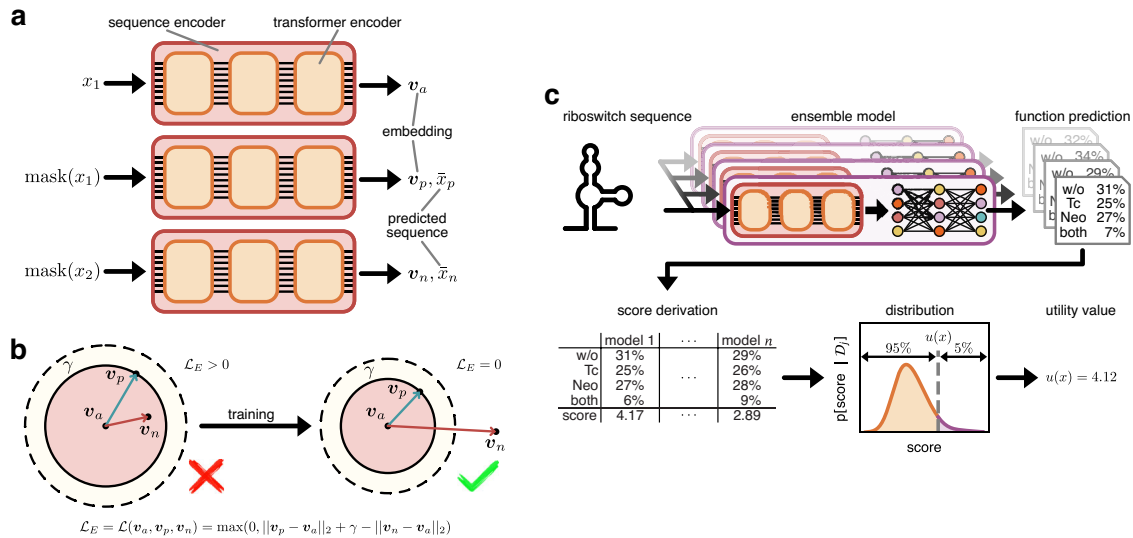


Figure 4. Pretraining and deep ensemble model. **(a)** Triplet network setup for pretraining of the sequence encoder. The same sequence encoder (here depicted three times) is applied to the sequences x_1 , the masked version of x_1 , and the masked version of x_2 . The output values are then considered by the loss function (Equation 2). **(b)** Visualization of the triplet margin loss. Training maximizes the embedding distance between distinct sequences. **(c)** Sequence-to-function neural network ensemble as Bayesian optimization utility function instantiating the UCB.

Given the pretrained sequence encoder, the second stage of training focuses on the sequence-to-function ensemble model. In particular, a fine-tuning approach is applied where each network in the ensemble is trained independently of each other on the same dataset. Despite the same training data, the random initialization of the parameters of the MLP ensures the diversity of the ensemble [48, 49].

As a surrogate model for Bayesian optimization, the trained ensemble predicts the empirical distribution of the score from which the utility of the sequence is derived (Fig. 4c), while the fine-tuning process is repeated from scratch for every Bayesian optimization iteration.

DBTL-cycle for NAND P1 stem design

We next apply the proposed batch Bayesian optimization framework to the design of the TNNAND8 P1 stem by combining it with an *in vivo* riboswitch screening. This combination allows for a functionality-driven design of the P1 stem while using the individual construct measurements required for accurate design and simultaneously tackling the sequence space complexity. In particular, our method comprises three stages: (i) the design of a riboswitch sequence pool, (ii) the screening of the sequence pool for dataset creation, and (iii) our Bayesian optimization framework for the efficient and effective design of riboswitch sequences. In the initial step (Fig. 5a), the sequence pool is created by using the sequence of TNNAND8 as the basis for the generation of a pool of partially randomized oligonucleotides (randomization of P1 stem). By employing fluorescence-activated cell sorting (Fig. 5b), we enrich sequences that demonstrate sufficient dynamic range between the absence and the presence of both ligands. Only these sequences are taken into account by Bayesian optimization. As depicted in Fig. 5c, this instantiates a DBTL cycle. After multiple rounds, this approach results in riboswitch sequences maximizing the desired NAND functionality. We describe each of these steps below and provide a flowchart representation in [Supplementary Fig. S2](#).

The screening process focuses on enriching riboswitches with a high dynamic range between the presence and the absence of both ligands. With the goal of increasing sequence diversity without losing pool coverage, we derive four distinct sequence pool designs of TNNAND8. Of note, P1 stems of the tetracycline aptamer conferring improved regulation are likely to contain certain closing base pair combinations below the binding pocket (C–G, C–G/C–G, C–G/G–C) [63]. Therefore, each pool is designed to contain a five base pair long randomized sequence within the tetracycline P1 stem and either no additional closing base pair, or one of the aforementioned closing base pair combinations (Fig. 5a, Pools T1–4 from left to right). We separately transformed *S. cerevisiae* cultures with the four resulting N10 pools ($4^{10} = 1\,048\,576$ variants per N10 pool; transformation efficiency per pool $\sim 5 \times 10^6$ cfu/ μ g plasmid). Afterward, we combined the four cultures.

With the sorting range being defined on the basis of control plasmids, we first sorted the combined N10 pools for low fluorescence values (0%–10%) in the presence of both ligands to remove riboswitches without function (1.2 million sorted cells) (Fig. 5b). The sorted subpopulation was allowed to grow and regenerate before the next sorting step. For the second sorting we created equally sized gates dividing GFP values between 20%–100% into four different subpopulations. We did this to differentiate between candidates with lower and higher dynamic ranges to avoid enriching only high performing hybrid riboswitches that have abolished NAND logic. We based the amount of sorted cells per gate on the size of the respective subpopulation to maximize coverage of all contained sequence variants (20%–40% GFP: 360k cells; 40%–60% GFP: 36k cells; 60%–80% GFP: 10k cells; 80%–100% GFP: 10k cells). Afterward, each subpopulation was individually sorted in the presence of both ligands to further enrich candidates with low OFF state fluorescence (10k sorted cells each).

We again extracted and amplified the sequences to analyze them via NGS. In preparation for the computational framework, we created two datasets. Dataset 1 contains all sequences identified via NGS and will serve as the dataset for

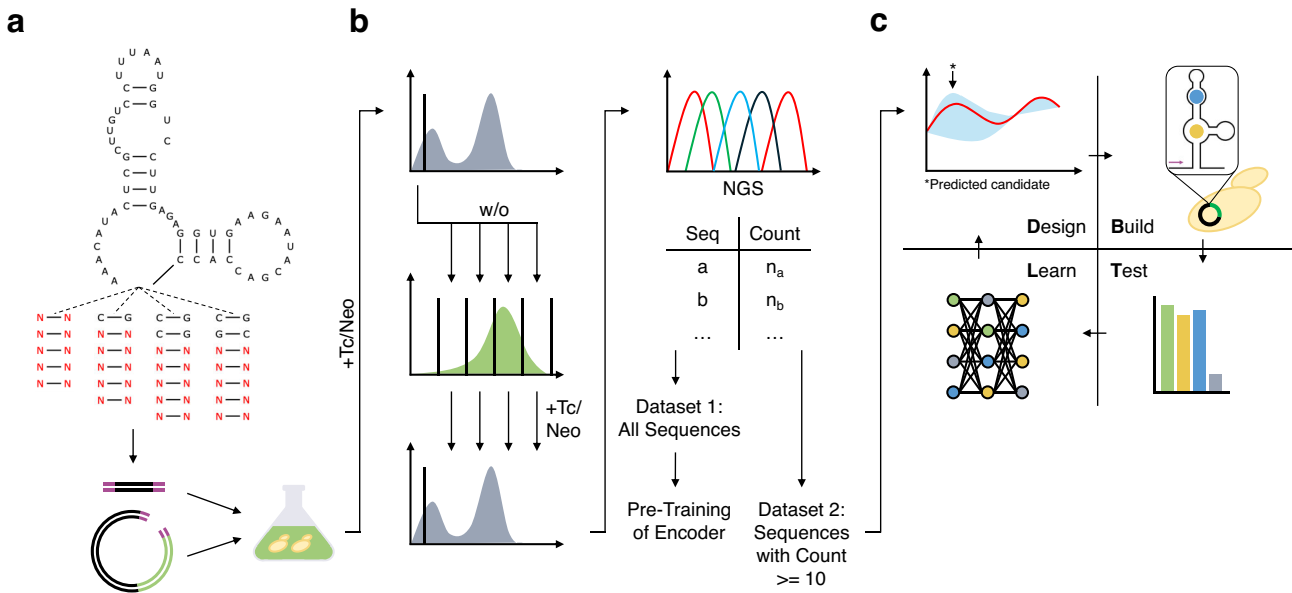


Figure 5. Overview on machine learning-guided NAND hybrid riboswitch design. **(a)** TNNAND8 P1 stem sequence pool design. Yeast libraries were generated through homologous recombination with the linearized vector pCBB06. **(b)** Sequence pool and dataset creation. Candidates with high dynamic range in the presence of both ligands were enriched through cell sorting. Four different gates were used for the ligand-free state sorting of candidates previously enriched for low reporter expression in the presence of both ligands to gain a better understanding of riboswitch behavior at different dynamic ranges. Afterward, each subpopulation was again enriched for candidates with low fluorescence in the presence of both ligands. NGS was used to determine the sequence space contained in each enriched subpopulation. For pretraining of the encoder, a first dataset containing all detected sequence variants was used. A second dataset only containing sequences discovered 10 times or more was used for model training. **(c)** DBTL cycle. Based on the initial training, the model predicted potentially interesting candidates for *in vivo* validation (Design). The respective constructs were cloned (Build), measured in yeast (Test), and the resulting data were used to train the model for the next iteration (Learn).

the pretraining of the sequence-encoder model. In contrast to this, dataset 2 will serve as the candidate sequence pool from which the batch Bayesian optimization selects the sequences for experimental characterization. To increase the experimental efficiency by removing sequencing artefacts, we required a read count of at least 10 for sequences to be part of dataset 2.

Application of Bayesian optimization to riboswitch design

The application of Bayesian optimization to the riboswitch sequence design requires the definition of a design objective, the scoring function. The scoring function characterizes the functionality of the riboswitch, which we define as the worst-case dynamic range with respect to NAND behavior. It assesses the dynamic range between the lowest expression level representing a Boolean ON state (ligand conditions w/o, Tc, and Neo) and the one representing the Boolean OFF state (ligand condition both). Formally, this score $S(\cdot)$ is defined as

$$S(x) = \frac{\min[\phi_x(\text{w/o}), \phi_x(\text{Tc}), \phi_x(\text{Neo})]}{\phi_x(\text{both})},$$

where $\phi_x(\cdot)$ is the expression level of sequence x for some ligand and condition.

For the sequence design framework, we use a batch size of $m = 16$ sequences, an ensemble size of $n = 100$ models, and configure the acquisition function UCB to bound 95% of predicted scores by setting $\alpha = 0.05$. For pretraining the sequence-encoder, dataset 1 consists of 36 925 riboswitch sequences. The candidate sequence pool, from which we select the sequences during our optimization, is dataset 2 and has a size of 3256 sequences (see Fig. 6 for sequence space visualization).

For the selection of the 34 initial riboswitches to characterize, we employed a semi-random approach. We rationally selected the majority of riboswitches based on their potential function derived from enrichment during screening. To also ensure sequence space coverage, we supplemented the initial riboswitches with candidates selected based on sequence similarity derived from the Levenshtein distance. All chosen candidates (TNNAND11-45; TNNAND29 was not considered further due to a later discovered sequence overlap with TNNAND45), despite not being sorted for a specific logic gate pattern, retained NAND logic gate behavior (Supplementary Fig. S3). To us, these results clearly indicate that the TNNAND8 communication module specifically enforces NAND logic gate behavior.

Starting the first round of our Bayesian optimization technique, the ensemble regression model is trained on the 34 sequences and the corresponding expression levels. For all sequences in the sequence pool, we evaluate the UCB and select the sequence with the highest utility value. The next iterations of this round continue by augmenting the current knowledge with the estimated expression levels of the selected sequence (the KB step), training the regression model, and selecting the sequence with the highest UCB. Upon selection of all sequences for the current batch, we continue with the *in vivo* evaluation, the build and test steps of the DBTL cycle in Fig. 5c.

In this first round of batch Bayesian optimization (TNNAND46-61) we found 11 riboswitch sequences that performed better than the previous average and three that outperformed the previous best. Among those 16 sequences, there are also four sequences without NAND functionality (Supplementary Fig. S4). Noticeably, their stems have at most

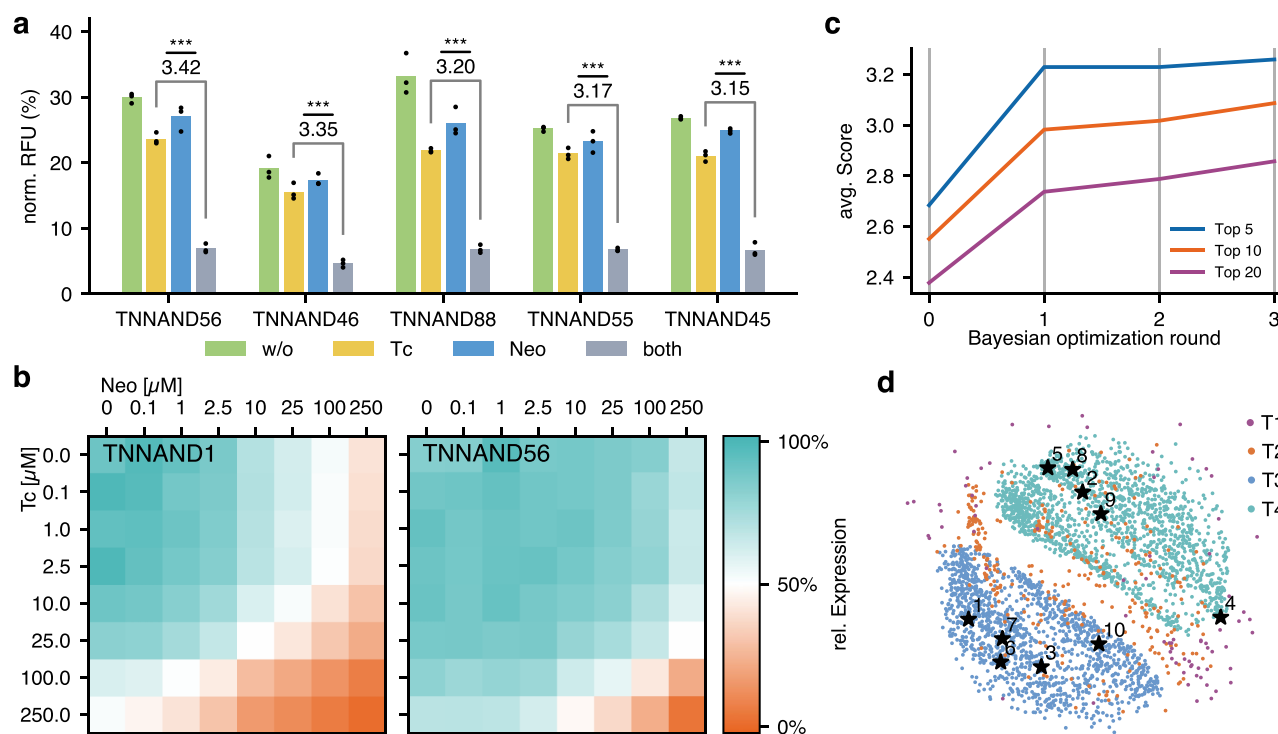


Figure 6. NAND riboswitch design results. **(a)** Relative fluorescence and scores of the Top five NAND riboswitches TNNAND56 (score 3.42), TNNAND46 (score 3.35), TNNAND88 (score 3.20), TNNAND55 (score 3.17), and TNNAND45 (score 3.15), ordered by descending performance quantified by the worst-case fold-change score. *P*-Values corresponding to worst-case dynamic range are indicated (see the “Methods” section). **(b)** Visualization of riboswitch dose dependency. TNNAND56 significantly downregulates expression only in the presence of both ligands, exemplifying the NAND behavior. In contrast, TNNAND1 exhibits regulation already in the presence of a single ligand. Color coding refers to the relative expression of each construct (min-max normalization per construct). **(c)** The average score of the Top 5, Top 10, and Top 20 sequences in each round of Bayesian optimization. **(d)** Sequence space visualization of the candidate sequence pool (dataset 2) by multidimensional scaling of pairwise Levenshtein distances. Clustering of sequences corresponding to pools T3 and T4 is likely due to sequences from these pools being the most prevalent ones. The Top 10 riboswitches distribute in the sequence space and only form vague clusters.

two canonical base pairs. Guided by those insights and the rationale of the riboswitch mechanism [12, 13], we consider the absence of base pairs in the stem causal for the absence of any switching behavior. Therefore, we removed sequences featuring <3 base pairs (including G–U pairs) in the stem from the sequence pool, to focus exploration on promising candidates. This reduces the size of the sequence pool by 2.4% to 3177 sequences.

Including the expression levels of the newly evaluated sequences into our dataset extends it to 50 entries. Rounds two and three of our Bayesian optimization proposed 32 further sequences. However, the *in vivo* characterization revealed that these rounds did not yield a new overall best NAND riboswitch. In addition, the predicted utilities of yet unknown riboswitches decreased, wherefore we terminated optimization with only 82 riboswitches being evaluated *in vivo*.

Considering the performance of the newly designed riboswitch sequences, Fig. 6a presents for each ligand combination the average expression levels regulated by the five best NAND riboswitches. Noticeably, the expression levels representing Boolean ON states (i.e. for input combinations w/o, Tc, and Neo) are well separated from the ones representing OFF states (i.e. for input combination both). In addition, the lowest ON state is on average only 22.1% below the highest, with TNNAND88 having the highest deviation of 34.2%. In combination with the average worst-case dynamic range

of 3.26 among our Top 5, these riboswitches implement near digital NAND logic *in vivo*. Compared to TNNAND8 (score 1.7), the Top 5 candidates outperform the initial construct score wise, with our method improving the worst-case dynamic range by two-fold for the best candidate TNNAND56 (score 3.42).

Figure 6b compares the dose-response behavior of TNNAND1, the initial hybrid riboswitch, and TNNAND56, the computationally designed NAND hybrid riboswitch. TNNAND1 partially blocks expression already when a single ligand is present (at concentrations > 25 μ M). Addition of the second ligand, even at lower concentrations, is sufficient to downregulate expression almost entirely. TNNAND56 however exhibits strong downregulation of expression only if both ligands are present to a significant extent (100 μ M or higher). Decreasing the abundance of one of the ligands increases expression levels immediately.

Investigating the convergence of the proposed technique, Fig. 6c highlights that the presented method continuously improves the average scores of the best riboswitches (Top 5, Top 10, Top 20) with each round of Bayesian optimization. While the 16 sequences selected in round 1 led to the largest improvements, the increases in later rounds are mostly of incremental nature. However, especially with respect to the Top 10 and the Top 20, the NAND score increase between round 1 and round 3 is still significant. Considering the design sequence space (Fig. 6d), we furthermore observe that the Top 10

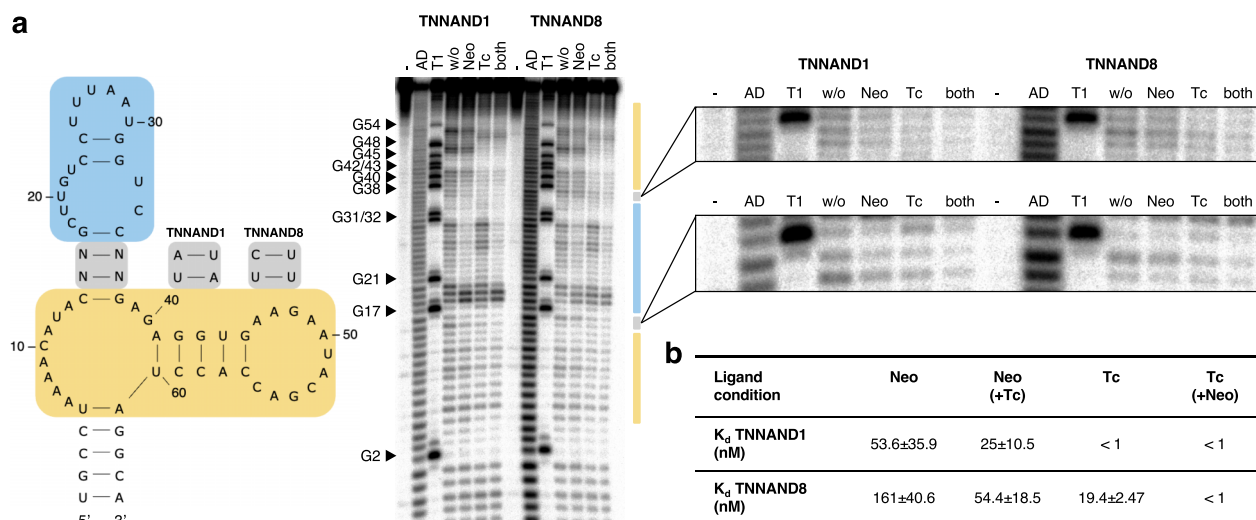


Figure 7. Structural and thermodynamic characterization of TNNAND1 and TNNAND8. **(a)** Two-dimensional structure prediction of the TNNAND1 and TNNAND8 hybrid riboswitches. The binding pockets for tetracycline and neomycin are highlighted in yellow and blue, respectively. The cleavage pattern of 32^P -labeled RNA (TNNAND1: left side, TNNAND8: right side) is shown after a 72-h in-line reaction under different ligand conditions (w/o: no ligand, Neo: 1 mM neomycin, Tc: 1 mM tetracycline, both: 1mM of each ligand). Size markers were generated through alkaline digestion (AD) and guanine residues were identified using T1 RNase digestion (T1). An untreated control (–) was prepared alongside the reactions. **(b)** K_d values were determined for TNNAND1 and TNNAND8 using ITC measurements (displayed in [Supplementary Fig. S5](#)). After titration of the first ligand (tetracycline or neomycin), the same RNA sample was titrated with the second ligand to measure binding affinity for the second ligand after binding of the first. Replicates were performed for each ITC measurement.

riboswitches form only vague clusters and instead distribute over the sequence space.

Structural and thermodynamic analysis of NAND hybrid riboswitches

Having identified candidates capable of implementing near digital NAND behavior, we next investigated the structural and thermodynamic characteristics of NAND hybrid riboswitches. We observed that all candidates discovered during the second screening implement NAND behavior, even before application of our sequence design framework. These constructs share the same communication module derived from the precursor NAND riboswitch TNNAND8, whose sequence differs from the original TNNAND1 riboswitch by only two nucleotide positions (A16C/A37U). Therefore, we compared TNNAND1 and TNNAND8 regarding secondary structure formation and binding affinity through in-line probing and ITC.

The results of the structural probing are depicted in [Fig. 7a](#). We confirmed the binding pockets for neomycin and tetracycline based on prior structure analysis of the individual parental riboswitches (tetracycline: yellow; neomycin: blue) [62, 70]. TNNAND1 and TNNAND8 both exhibit formation of the individual binding pockets in the presence of their respective ligands. In both constructs only the binding pocket of the present ligand is stabilized, while the unbound binding pocket remains in a less structured state. Addition of the second ligand then leads to stabilization of the entire construct. The cleavage patterns present no alternative structure formation of the NAND construct TNNAND8 in comparison to TNNAND1. However, signal intensities in the region of the communication module indicate increased flexibility at the positions C16, U36, and U37 of TNNAND8 in comparison to TNNAND1 for the ON states (w/o, Tc, Neo). In the

presence of both ligands, cleavage patterns of TNNAND1 and TNNAND8 indicate similar degrees of structural stability.

Binding affinity assays revealed that K_d values of TNNAND8 are worse in the presence of only one ligand when compared to TNNAND1 (see [Fig. 7b](#) and [Supplementary Fig. S5](#)). Before binding of tetracycline, TNNAND8 reaches a K_d of 161 nM in the presence of neomycin, whereas TNNAND1 achieves a K_d of 53.6 nM. Binding assays performed with neomycin after addition of tetracycline to the RNA revealed improved K_d values of 54.4 nM for TNNAND8 and 25 nM for TNNAND1. Binding of tetracycline in the presence of neomycin occurs with high affinity for both constructs, with K_d values dropping below the threshold of the utilized measurement device (<1 nM). Binding of tetracycline without prior addition of neomycin occurs at worse affinity for TNNAND8 when compared to TNNAND1, with only the K_d of TNNAND1 dropping below the measurement threshold.

Sequence characteristics of NAND hybrid riboswitches

Following the insights on the communication module, we here investigate the relationship between sequence and function with respect to the P1 stem. We do so by considering the expression profiles of the 82 riboswitches characterized during P1 stem design.

[Figure 8a](#) gives rise to the nucleotide frequency per stem position for the Top 10 sequences according to the NAND score. While for most positions each nucleotide appears at least once, the preference for sequences from pool T3 (5 out of 10) and T4 (4 out of 10) in the Top 10 is evident at position 7, as only a single sequence corresponds to pool T2 (1 out of 10) with a stem of length six. At position 6, nucleotides G and C are almost equally frequent, again reflecting the ratio of pools in the Top 10. Considering pool T1, featuring the shortest stem of length five and no predefined base pairs, there

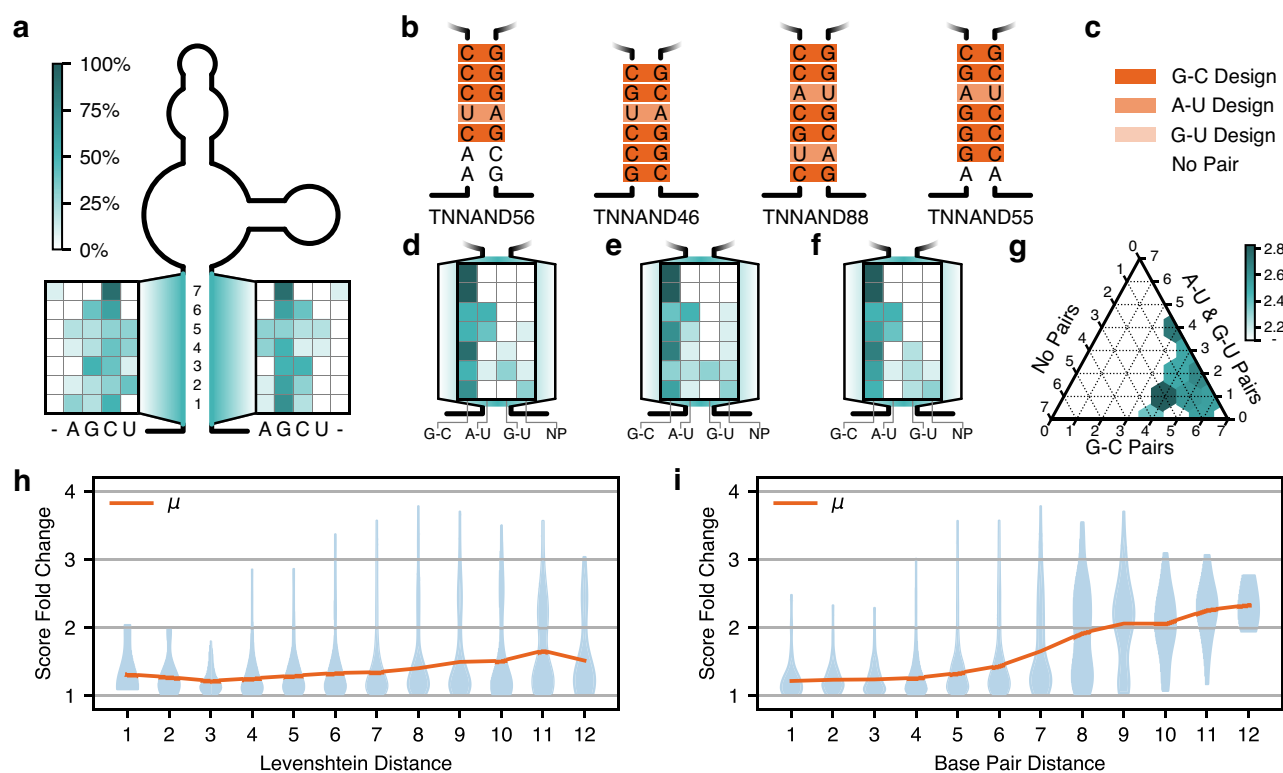


Figure 8. Sequence insights into riboswitch design. (a) Relative frequency of nucleotides of Top 10 riboswitch sequences, where – indicates the absence of a base through a shorter stem. (b) Sequences of Top four riboswitches designed in this work with highlighting of base pairs following the legend in panel (c). Relative frequency of base pairs in Top 10 of the riboswitches with respect to the score (d), high expression in the “w/o” case (e), and low expression in the “both” case (f). (g) Average scores of Top 40 riboswitches in dependence to the number of base pairs, where we differentiate between no base pairing (NP), G–C base pairing (G–C pairs), and A–U or G–U base pairing (A–U and G–U pairs). (h) Riboswitch sequence robustness analysis on the basis of the distribution of the fold change in score (lower is better) in dependence to the Levenshtein distance between sequences, with substitutions, insertions, and deletions weighted equally. Small score fold change means almost identical scores. (i) The distribution of the fold change in score in relation to a base pair distance, where the distance between two distinct base pairs is 1 and between a base pair and no base pair is 2.

is no sequence present in the Top 10 and only three—in general nonfunctional—sequences from T1 have been evaluated in total. Positions 1 to 5 are not predefined by the pools and present a more heterogeneous choice of nucleotides. In general, G is the most prominent nucleotide on the right side, with the complementary C occurring slightly less often on the left.

Figure 8b displays the P1 stem of the Top four riboswitches with position-wise base pairs highlighted (legend in Fig. 8c). Notably, the best performing riboswitch TNNAND56 features only five base pairs, all of which are in the upper part. The other top riboswitches feature up to seven. In addition, there is a strong prevalence for G–C base pairs (19 in total) over A–U base pairs (5 in total). Here, TNNAND56 again diverges from the others in featuring three G–C base pairs in the upper part of the stem.

Considering the base pairing of the 10 NAND riboswitches with the highest score (see Fig. 8d), G–C base pairs are most prevalent. A–U pairs occur only at positions 2, 4, and 5. The noncanonical G–U pair and the absence of base pairs is mainly present at the lower part of the stem. Figure 8e and f continue this evaluation by focusing on the 10 riboswitches with highest expression levels in the “w/o” case (panel e) and the 10 with lowest expression levels in the “both” case (panel f). Comparing panels (e) and (f), one observes that the base pairing patterns differ. The increased absence of base pairs in the w/o case indicates that high expression is best achieved by weakening the base pairing. Contrary to that, we observe

frequent use of G–C base pairs in sequences with the least leakiness in the “both” case.

Reconsidering the score, Fig. 8g visualizes the average score in dependence to the number of base pairs per type for the 40 best NAND riboswitches. All riboswitches have at most three positions without base pairing. On average, the configurations that perform best have four pairs of type G–C and one of type A–U or G–U, while riboswitches with seven base pairs perform better than average.

An important aspect for sequence design and the predictability of functionality is the robustness of the construct with respect to altering single nucleotides. Figure 8h presents the distribution of the fold change in score between riboswitches with different Levenshtein distances. The score fold change is s_1/s_2 if $s_1 \geq s_2$ and otherwise s_2/s_1 , with $s_1 = S(x_1)$ and $s_2 = S(x_2)$, and characterizes the deviation in score. Focusing on the mean first, we observe a slight increase in score fold change for increasing Levenshtein distance. Taking the distribution information into account, we observe that distributions for different Levenshtein distances are quite similar, while already a Levenshtein distance of 1 can lead to a two-fold change in NAND score. As the number of base pairs and their type is an important quantity (see Fig. 8d and e for example), we additionally introduce a Base pair distance (see Fig. 8i). The base pair distance weights the distance between two distinct base pairs with 1 and between no base pair and a base pair with 2. The average score fold change in panel

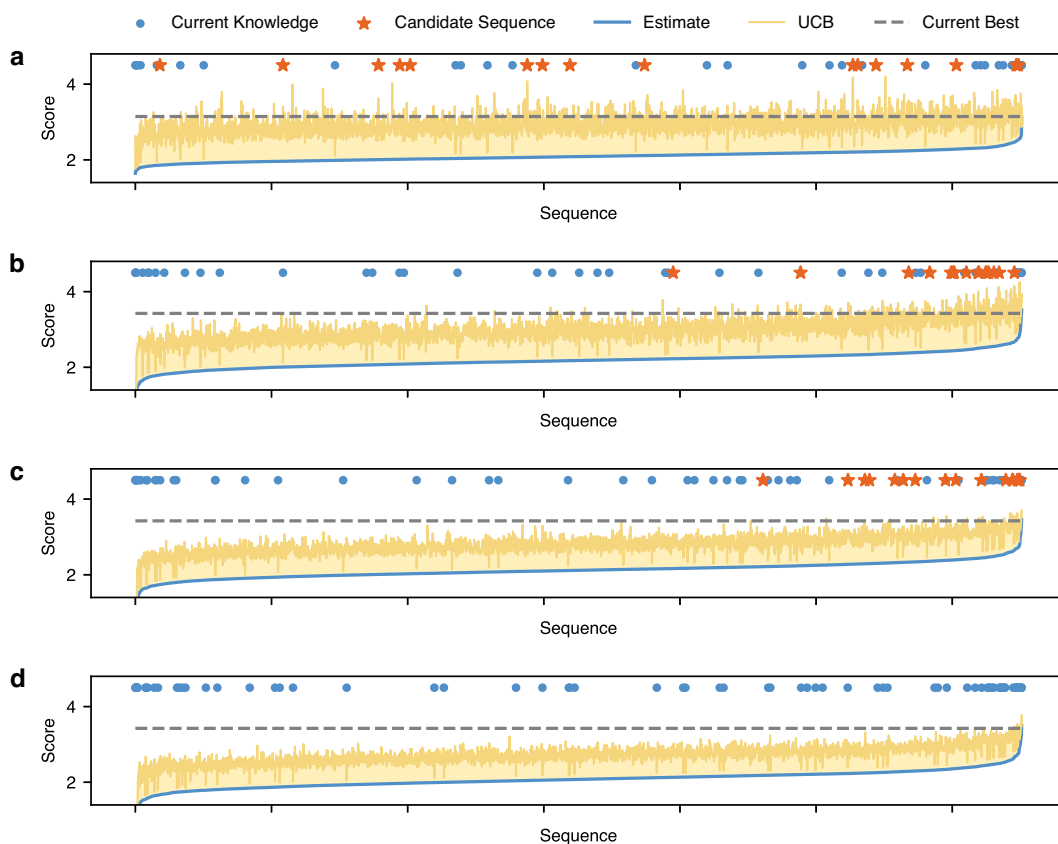


Figure 9. Batch Bayesian optimization rounds. Visualizations of the model's predictions ($\hat{x}$) (blue line) and the corresponding UCB $u(x)$ (UCB, yellow area) for each sequence in the candidate sequence pool on the basis of the knowledge at the start of round 1 (a), 2 (b), and 3 (c), with panel (d) representing the state for an unexecuted fourth round. In addition, markers indicate current knowledge (blue points) and proposed sequences to evaluate (orange stars). Sequences are ordered per round by ascending predicted score. Deviations between UCB peaks and selected candidate sequences result from different model states for visualization and sequence selection.

(i) almost monotonically increases and spans a much larger range in comparison to panel (h). In addition, the distributions differ more significantly, indicating the importance of base pairing for behavioral similarity.

Analysis of the Bayesian optimizer

With having covered the evolution of top riboswitch performance over Bayesian optimization rounds, we here focus on providing further insights into the selection process. Figure 9 visualizes the model's view of the sequence space at the beginning of each Bayesian optimization round [round 4 corresponding to panel (d) is not executed]. Starting with the model's knowledge at the beginning of the first round (panel a), we see that the characterized riboswitches span the whole performance range from almost nonfunctional to NAND behavior. The model's uncertainty vanishes for characterized riboswitches (downward spikes in UCB) while otherwise the distance between UCB (representing the 0.95% quantile) and mean estimate is evident for the large uncertainty within the model.

As in general the UCB values are high, the riboswitches selected are distributed over the whole sequence space, including candidates with low estimated performance but high uncertainty (exemplary for exploration), and candidates with lower uncertainty but higher performance estimates (exemplary for exploitation). According to the definition of the acquisition function, the selected candidate sequences should correspond

to peaks in the UCB utility function. However, this does not align with our observations in the panels. This is because the presented UCB is representative for the model's knowledge at the beginning of the Bayesian optimization round. This is not identical to the model's belief in later iterations when the other candidate riboswitches are selected. Causal to this is the KB approach, which updates the model's belief in each iteration to reflect the knowledge gained through the incorporation of the previously selected riboswitches.

Continuing with round 2 (see Fig. 9b), we observe a general drop in uncertainty (deviation between UCB and mean) and a preference for riboswitches with higher estimated performance in comparison to round 1 [panel (a) for reference]. Indeed, the information gained through the characterization of the riboswitches selected in round 1 refines the predictions, also in the realm of sequences with low estimated performance. The decreased predictive uncertainty reduces their selection probability due to the combination of low estimated performance and medium uncertainty levels. At the other end of the performance spectrum, uncertainty is causal for candidate selection, as the difference in the expected performance of predicted top performers decreases. With respect to the diversity of selected riboswitches, the model's knowledge did lead to preferring exploitation, with the exception of some candidates being chosen for the purpose of exploration.

Round 3 manifests the trend of decreasing uncertainty and the selection of candidates with higher estimated performance

(see Fig. 9c). Especially in the range of riboswitches with medium estimated score, single peaks in uncertainty are causal for the selection of a riboswitch. However, we also do notice that in most cases the distance between selected candidates is larger than in round 2. This indicates the success of the KB approach in reducing uncertainty in the proximity of riboswitches already part of the current batch. Besides, the selected riboswitches again indicate the trade-off between exploration and exploitation by considering sequences with diverse estimated performance. This is in contrast to greedy optimization, which focuses only on the sequences with the highest estimated performance.

After three rounds of batch Bayesian optimization, the model's uncertainty decreased significantly (see Fig. 9d) and the UCB is in general below the already identified best scoring riboswitch. This indicates the vanishing chance of identifying riboswitches with superior performance. The scores of the evaluated riboswitches span from 0.91, representing an unfunctional riboswitch, to 3.42 in the case of TNNAND56. From the 48 riboswitch sequences selected by batch Bayesian optimization only 14 have a score below 1.70 (score of TNNAND8), while 15 switches exceed this score by at least 50%.

In order to assess the applicability of our deep batch Bayesian optimization to other sequence design tasks, we ran multiple campaigns with our framework on an *in silico* benchmark for promoter design [71], which has a candidate sequence space of over 700 000 sequences (see [Supplementary Fig. S6](#)). We compared our framework to a round based greedy optimization policy based on the state-of-the-art reference neural network for this dataset. The objective was to maximize the expression, while the parametrization of Bayesian optimization stayed the same otherwise. Our sequence design framework was readily applicable to the promoter design after adapting the ensemble model to the promoter task. Considering the performance (multiple independent campaigns are considered), our deep batch Bayesian optimization framework outperforms the reference neural network policy in designing promoter sequences ([Supplementary Fig. S6](#)).

Discussion

RNA-based regulators such as synthetic riboswitches offer enormous potential for the design of complex genetic circuits. Despite this, riboswitches are difficult to engineer due to their sensitivity to sequence changes, with even single nucleotide exchanges having a major impact on regulated expression levels. Because sequence–function relationships are often unpredictable, riboswitch design in many cases remains dependent on time-consuming empirical methods. Machine learning approaches have attempted to bridge this gap by combining high-throughput wet lab data generation with sequence-to-function surrogate models. However, these studies have focused exclusively on the generation of single-input RNA devices with maximized dynamic range [24, 25, 63]. Our hybrid riboswitch architecture surpasses these devices in complexity by allowing the use of two different inputs to simultaneously influence the switching behavior, thereby enabling the implementation of Boolean logic. This complexity is further enhanced by the fact that hybrid riboswitch function is based solely on structure stabilization induced by small-molecule triggers, unlike other riboswitch architectures that

rely on strand displacement to facilitate regulation [24, 25]. Until now, the design of hybrid riboswitches has focused solely on maximizing dynamic range to create highly efficient regulators that take advantage of the additional stability provided by a second binding pocket [14]. Most of these rationally designed constructs already exhibit a high dynamic range in the presence of a single ligand, thus achieving NOR behavior without the need for extensive optimization. Here, we investigated logic behavior that requires conditional functionality of the construct only in the presence of both ligands, with the specific goal of achieving near digital NAND logic behavior.

Most dual-input riboswitch architectures attempting to implement Boolean logic rely on separate insertion of two different devices into the same messenger RNA. While these tandem riboswitches can be created from the use of two different constructs utilizing the same mechanism of regulation, like the roadblock mechanism in *S. cerevisiae* [72, 22], simultaneous use of different regulation mechanisms is also conceivable [10, 73]. Tandem riboswitches following Boolean logic have not only been engineered synthetically but also been discovered in nature [74]. For example, a tandem construct containing aptamers for guanine and 5-phosphoribosyl-1-pyrophosphate discovered in bacteria was shown to achieve IMPLY logic by influencing a shared expression platform (terminator stem) through both binding pockets [75]. The fact that no natural riboswitch implementing NAND logic has been discovered to date is viewed as an indicator that interactions between both binding pockets are a requirement for dual-input riboswitch architectures to achieve complex Boolean functions [15]. Previously published synthetic riboswitches implementing NAND behavior circumvent this by using a self-cleaving ribozyme, a catalytically active RNA sequence, as a shared expression platform for two different aptamer domains [16, 17]. These designs make use of the modular nature of ribozymes by replacing two stem-loop structures essential to their function with aptamer domains that stabilize only in the bound state, thus allowing ribozyme activity only in ligand presence. However, the spatial separation between both binding pockets means that no direct interaction is utilized for NAND behavior, as each aptamer domain separately functions as a NOT gate. In comparison, our design utilizes direct interactions between both binding pockets to facilitate logic behavior, thus encoding NAND logic directly into a single RNA structure without the need for a separate expression platform.

Our tetracycline–neomycin hybrid riboswitch accurately emulates NAND logic, showing relevant downregulation of reporter expression only, and exclusively, in the presence of both ligands. We have validated this behavior *in vivo* in a dose-dependent manner. Additional *in vitro* characterization used for comparison with the original riboswitch TNNAND1 revealed increased flexibility of the communication module conveying NAND logic in the presence of a single ligand. The reduced stability of the communication module potentially affects the ability of NAND constructs to form a stable roadblock *in vivo*, thus requiring simultaneous binding of both ligands to efficiently terminate ribosomal scanning. In addition, ITC measurements revealed decreased binding affinity of a NAND riboswitch for individual ligands. Interestingly, both the original TNNAND1 construct and the NAND riboswitch exhibit increased binding affinity for the second ligand (tetracycline or neomycin) after binding of the first. This effect, which is more pronounced for the NAND construct, indicates that both riboswitches utilize cooperative binding.

This means that each binding pocket, in its bound state, increases the affinity of the other binding pocket for its respective ligand. Since we observe this effect independently of logic behavior, cooperative binding may be more generally associated with the hybrid riboswitch architecture and potentially applicable to other binding pocket combinations.

We choose Bayesian optimization to guide riboswitch design. Previous works have used classical design methods like directed evolution [18–22] or machine learning based approaches for the design of regulatory RNA devices [24, 25, 76, 77]. The collaborative work [24, 25] combines high throughput cell sorting and sequencing with deep learning-based sequence-to-function models to predict RNA toehold-switch function. Their optimization pipeline uses the trained model to optimize the input RNA sequence via gradient ascent with respect to target expression values in the bound and unbound state. In comparison, our batch Bayesian optimization based framework employs probabilistic sequence-to-function modeling, additionally giving rise to the predictive uncertainty. This estimate of predictive accuracy is essential to trade off exploration and exploitation in data scarce scenarios. Also a generative approach to RNA aptamers on the basis of structure-to-sequence models has been proposed [77]. This was exemplified by generating structurally similar but sequence-wise dissimilar variants of light-up aptamers. Our framework directly focuses on the functional behavior of the riboswitch and provides means to maximize the performance with respect to a functionality score.

Bayesian optimization itself is already employed to guide sequence design in experimental design campaigns like protein engineering [34, 78] or small molecule design [37]. These works use greedy approaches, stochastic acquisition functions, or approximate joint uncertainty maximization for batch generation and significantly smaller ensembles [34] or single neural network techniques such as Laplace approximation, Monte Carlo dropout, or Gaussian process heads [37, 34, 78] for probabilistic modeling. Our method goes beyond recent works by explicit batch creation and overcoming distributional assumptions. We explicitly condition the selection of riboswitches during batch generation on previously selected candidates through the use of KB to prevent the selection of too similar riboswitches and to ensure batch diversity. In addition, we account for sequences as inputs and arbitrary empirical distributions by using a large ensemble neural network and introducing the quantile-based UCB acquisition function. This novel combination of KB and deep ensemble neural networks as surrogate sequence-to-function model presents a Gaussian-process-free alternative for Bayesian optimization, overcoming the need of intermediate sequence embeddings and making assumptions on the predictive distribution superfluous.

Using the deep batch Bayesian optimization, we were able to double the performance of the optimized candidate TNNAND56 in comparison to the initially discovered NAND predecessor TNNAND8. By having a total sample size of only 82 candidates validated *in vivo*, our framework shows robust performance in this low-throughput regime. We originally considered utilizing larger pool sizes in a high-throughput screening for this task. However, the complexity of designing a multistate device that precisely emulates Boolean logic behavior makes it increasingly difficult to generate datasets containing the required information for all four states when working with larger pools. A limiting factor for high-throughput

screenings is the number of cells that can be sorted within the time frame permitted by the experimental parameters. Therefore, pool sizes that reach the order of cell count result in loss of information, even when screening for a single state [79].

We complement our results with an *in silico* benchmark for promoter design [71] to allow for general statements on the performance of our design framework. The benchmark showed that our sequence design method is easily adaptable to the new design task and demonstrated superior performance in comparison to the state-of-the-art single-neural-network-based optimization strategy. For our design framework, this performance comes at the cost of a significant increase in computational complexity, which is proportional to the product of ensemble size and batch size. However, the time for *in vivo* characterization still significantly exceeds the time of *in silico* batch creation, emphasizing the importance of high quality sequence selection.

Future extensions of our framework can integrate further knowledge into the modeling, for example by taking structural information into account [80–85]. In the context of Bayesian optimization, the integration of means to trade-off data quality and quantity [26, 27] is an orthogonal aspect to the previous and opens up a new dimension, which is promising for its application to RNA design but also to other sequence design tasks. Here, it is important to emphasize that the method presented can be adapted to other constructs beyond NAND hybrid riboswitches. By adjusting the *in vivo* screening and defining the sequence constraints and scoring objective for the *in silico* process, our framework can be extended to general 5' UTR design and adapted to other sequence design tasks for biological engineering. This includes the engineering of promoters, terminators, or enhancer sequences but also protein engineering when considered in nucleic acid representation (DNA or RNA). As such, our method can complement experiment-driven approaches by allowing fine grained adaptation of functionality through model-based and score-driven design, including constructs sensitive to single nucleotide changes. In addition, the novel NAND riboswitch we present has interesting implications for future riboswitch design. By utilizing our framework in conjunction with other binding pocket combinations, e.g. for metabolite sensing, a larger variety of compact riboswitches emulating NAND logic can be generated in a time efficient manner. This allows for more advanced circuit design at a low metabolic burden.

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Supplementary data

Supplementary data is available at NAR online.

Conflict of interest

None declared.

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Data availability

All data are provided in this article and the online supplementary material. Supplementary figures and construct/pool sequences are contained in the supplementary file. NGS Data are available at <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE293990>. Code is available at Github via the DOI [10.5281/zenodo.15185168](https://doi.org/10.5281/zenodo.15185168) [86].

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